Week 3 - Core - Worksheet | AI Solution Design & Development - Designing a Solution

Strand Self-Check

- **Rate yourself against this week's success criteria: Can do / Partially / Not yet. What would you practise to move up a level?**

Can do ☐ Partially ☐ Not yet ☐ To practise:

_____

Name: _____ **Class:** _____ **Date:** _____

Scenario: the clinic appointment helper - typed symptoms in, a suggested appointment type and a plain-language summary out.

1. Write, in one line, what a finished result of your chosen task should look like.

_____

_____

2. Design the workflow: for each step write its input and output, and mark the AI tool step.

Step	Input	Output	AI tool step?
1			
2			
3			

3. What does your prepare step check or remove before the AI step (including any personal data)?

_____

_____

4. Run it, then explain the result by tracing the final output back to the step that produced it.

_____

_____

5. My reflection: the step that mattered most in completing my task was... because...

_____

_____

Week 3 - Lab - Worksheet | AI Solution Design & Development - Build, Test, Improve

Name: _____ **Class:** _____ **Date:** _____

Scenario: the clinic appointment helper - make it reliable across more than one input, and combine a second tool and data source.

1. Run your Core workflow on a fresh input. Judge the output against your task. What is the weakest step?

_____

_____

2. Change only that one step. Record the result before and after, and the downstream output that changed.

Step changed	Before	After

3. Add a second tool step and a second data source. What does each add?

_____

_____

4. Run the whole solution on two different inputs. Did it complete the task both times? Verify each output.

_____

_____

5. My reflection: the step or data source I changed that improved my solution most was... and I know because...

_____

_____

Week 4 - Core - Worksheet | Critical Evaluation & Informed Interaction - Judging & Improving Outputs

Strand Self-Check

- **Rate yourself against this week's success criteria: Can do / Partially / Not yet. What would you practise to move up a level?**

Can do ☐ Partially ☐ Not yet ☐ To practise:

_____

Name: _____ **Class:** _____ **Date:** _____

Scenario: three AI translations of the same visitor notice - judge them against accuracy, fairness and usefulness, with evidence.

1. For one output, name a strength and a weakness against each criterion, with the exact evidence.

Criterion	Strength (+ evidence)	Weakness (+ evidence)
Accuracy		
Fairness		
Usefulness		

2. Trace the deciding weakness to a component. Was it the data, the prompt, or the tool? How do you know?

_____

_____

3. Explainability: could this system tell an affected person how it produced the translation? What could it show, and what stays hidden?

_____

_____

4. Think about it: could an output be perfectly accurate and still have a serious weakness? Give an example.

_____

_____

5. My reflection: the strength I almost missed was... , and the weakness that surprised me was...

_____

_____

Week 4 - Lab - Worksheet | Critical Evaluation & Informed Interaction - The Evaluation Clinic

Name: _____ **Class:** _____ **Date:** _____

Scenario: a messier batch of translations - rank them, test the front-runner across inputs, and fix the weakest component.

1. Rank the outputs best to worst. Write the deciding evidence and any tie-breaker beside each.

Rank	Output	Deciding evidence / tie-breaker
1		
2		
3		

2. Test the top output on a harder notice. Did it stay accurate? Note any false positive or false negative.

_____

_____

3. Take the weakest output. Trace the deciding weakness to a component (data / prompt / tool) and propose one fix.

_____

_____

4. Think about it: could fixing the prompt turn a fairness weakness into a strength, or would you need better data? How would you test it?

_____

_____

5. My reflection: my ranking changed when I noticed... , which taught me...

_____

_____

Week 5 - Core - Worksheet | Governance & Ethics - Locate the Risk in the System

Strand Self-Check

- **Rate yourself against this week's success criteria: Can do / Partially / Not yet. What would you practise to move up a level?**

Can do ☐ Partially ☐ Not yet ☐ To practise:

_____

Name: _____ **Class:** _____ **Date:** _____

Scenario: the school admissions shortlister scores applicants and produces a ranked list that staff act on: application data → scoring model → ranked list.

1. Find three concerns in the shortlister. For each, name its type, the component it comes from, the stage (1-6, if it is a bias), and a responsible measure.

Concern	Type (bias / privacy / misuse)	Component (data / model / prompt)	Stage (1-6)	Responsible measure

2. Choose one privacy concern above. Name one technical mechanism (data minimisation, anonymisation / pseudonymisation, or access control) that would protect people, and say why.

_____

_____

3. Think about it: an integrated service quietly reuses data given for one task to do a different task. Is that a privacy concern, a misuse concern, or both? How would your measure differ for each?

_____

_____

_____

4. My reflection: the ethical concern I judge most serious in the shortlister is...

_____

_____

Week 5 - Lab - Worksheet | Governance & Ethics - Measure the Risk and Govern It

Name: _____ **Class:** _____ **Date:** _____

Scenario: the admissions shortlister has been tested on applicants whose true strength is known. Use the results below to measure fairness and plan a responsible deployment.

1. Measure fairness. Enter the counts from the results pack; the false-negative rate and parity gap fill in automatically. (Rate = wrongly rejected / strong tested.)

Group	Strong applicants tested	Wrongly rejected (false	False-negative rate

		negatives)	
Group A			
Group B			

2. What does the gap mean for real applicants? Which group is the system less fair to, and which pipeline stage is the most likely source?

_____

_____

3. Propose a stage-targeted mitigation for the biggest concern - name the stage it changes and what you would do.

_____

_____

4. Write a deployment plan . Complete each step and name the accountable role.

Pilot

what runs alongside humans _____

accountable role _____

Monitor

what you track over time _____

accountable role _____

Incident response

what triggers a pause _____

accountable role _____

Redress

how a person appeals _____

accountable role _____

5. My reflection: the mitigation I would build in first is... , the person accountable for what remains is... , and I know the fix worked because the parity gap changed from ... to ...

_____

_____

Week 6 - Core - Worksheet | Integrated · all four strands - Bringing the Strands Together

Strand Self-Check

- **Rate yourself against this week's success criteria: Can do / Partially / Not yet. What would you practise to move up a level?**

Can do ☐ Partially ☐ Not yet ☐ To practise:

_____

Name: _____ **Class:** _____ **Date:** _____

Scenario: the Ne'ma-inspired food-surplus matcher - classify, match, draft - measured with a metric you define and compute.

1. Run ten item cards through Classify → Match → Draft. Tick each stage that matched the true key.

Item	Classify ok?	Match ok?	Draft ok?
1			
2			
3			
4			
5			

2. Compute your metric. Match accuracy = matches passing the rule sheet ÷ matches proposed, as a percentage.

	Matches passing	Matches proposed	Accuracy (%)
Before the constraint			
After adding the constraint			

_____

_____

3. Where did an early error propagate? Name the tool or interface that caused most misses.

_____

_____

4. Name one responsible-use choice you made (for example, a human gate or using no personal data).

_____

_____

5. My reflection: the tool or interface I trust most, and the one I would improve first, is... because...

_____

_____

Week 6 - Lab - Worksheet | Integrated · all four strands - Run It End-to-End

Name: _____ **Class:** _____ **Date:** _____

Scenario: inspect the surplus-item data, run the matcher end-to-end, measure accuracy, and adapt.

1. Inspect the dataset. Record one missing value, one wrong label, and whether the set is representative.

_____

_____

2. Run the full set and total your accuracy (items right at every stage) out of the total.

	Right at every stage	Total items	Accuracy (%)
Before the change			
After the change			

3. Name the tool or interface that failed most, and where an early error propagated.

_____

_____

4. Add one constraint (a second criterion or a risk rule). What did it change?

_____

_____

5. My reflection: the tool or interface that let my pipeline down most, and the change that helped, was... because...

_____

_____

Week 7 - Core - Worksheet | Signature Solution · Planning - Architect Your Signature Solution

Strand Self-Check

- **Rate yourself against this week's success criteria: Can do / Partially / Not yet. What would you practise to move up a level?**

Can do ☐ Partially ☐ Not yet ☐ To practise:

_____

Name: _____ **Class:** _____ **Date:** _____

Your brief: choose an authentic, non-personal problem and stakeholder, then architect an integrated solution for it.

1. Write your goal in one sentence: what defined problem will your integrated solution solve, and for whom?

_____

_____

2. List your components and draw the ordered structure (each tool's output feeds the next).

Data (non-personal)	Tool 1	Tool 2	Prompt for each

3. Data preparation: what is the data, how is it prepared and de-identified, and what does it exclude?

_____

_____

4. Set success criteria (a number), then name one risk, its mitigation, and the check step that catches it.

_____

_____

5. Start your system card: purpose, inputs, data & preparation, limitations, and who is accountable.

_____
_____
_____

Week 7 - Lab - Worksheet | Signature Solution · Planning - Prototype and Pressure-Test

Name: _____ **Class:** _____ **Date:** _____

Your brief: prototype your integrated solution by hand, then pressure-test it and design a fallback.

1. Walk one input through every tool. At each hand-off, note the output passed on and whether it is usable.

_____
_____
_____

2. At each check step, did the output meet your success criteria? Where did it pass or fail?

_____
_____

3. Weaken one component on purpose. Trace how far the failure spread before a check caught it.

Component weakened	Tools affected before a check caught it

4. Design a fallback path: the check that triggers it, the safer reroute, and who is accountable.

_____
_____

5. My reflection: the hand-off that held up best was... and the fix or fallback I added after pressure-testing was...

_____
_____

Week 8 - Core - Worksheet | Signature Solution - Build, Evaluate, Improve

Strand Self-Check

- **Rate yourself against this week's success criteria: Can do / Partially / Not yet. What would you practise to move up a level?**

Can do ☐ Partially ☐ Not yet ☐ To practise:

_____

Name: _____ **Class:** _____ **Date:** _____

Your brief: assemble your solution, evaluate it, and set aside a held-out test set for the final check.

1. List the inputs you will build and tune on, and the few you will hold out (do not tune on these).

Development inputs (build & tune)	Held-out inputs (test only at the end)

2. Assemble and run. Record the integrated output, then mark it against each success criterion and governance rule.

_____

_____

_____

3. Verify one claim in the output against a trusted source. What did you check, and against what?

_____

_____

4. Change one component to fix a missed criterion. Which one, and did the new output meet it?

_____

_____

5. My reflection: the one component I improved today was... because...

_____

_____

Week 8 - Lab - Worksheet | Signature Solution - Finish It, Then Prove It

Name: _____ **Class:** _____ **Date:** _____

Your brief: finish the system, test it several times including an edge case and a held-out input, and keep an evaluation log.

1. Keep an evaluation log. Add a row for each test.

Input (type)	Output	Criteria & rules met?	Component changed

2. Describe one edge case that broke a component, and how you made the system respond gracefully.

_____

_____

3. Run a held-out input the system was not tuned on. Did it still meet the criteria?

_____

_____

4. Across your logged tests, which improvement do you trust most, and what is your evidence?

_____

_____

5. My reflection: across my logged tests, the component improvement I trust most is... because...

_____

_____

Week 9 - Core - Worksheet | Signature Solution - Refining & Finishing the System

Strand Self-Check

- **Rate yourself against this week's success criteria: Can do / Partially / Not yet. What would you practise to move up a level?**

Can do ☐ Partially ☐ Not yet ☐ To practise:

_____

Name: _____ **Class:** _____ **Date:** _____

Your brief: make one traced change, re-run the chain, and measure the improvement on your held-out set.

1. Pick ONE component to change. Predict which hand-off will carry the effect.

_____

_____

2. Measure before and after on the held-out set. Accuracy = correct ÷ total.

	Correct	Total	Accuracy (%)
Before			
After			

_____

_____

3. Did the effect propagate through the hand-off you predicted? What happened downstream?

_____

_____

4. Risk check: who could be affected, how would you catch a mistake, and which governance rule applies?

_____

_____

5. My reflection: the one change that most improved my system was... , it propagated through... to... , and I know it helped because...

_____

_____

Week 9 - Lab - Worksheet | Signature Solution - Lock It and Rehearse the Demo

Name: _____ **Class:** _____ **Date:** _____

Your brief: lock a reproducible version and rehearse a defendable demo.

1. Run one final iteration. Confirm every component still meets the criteria, then version-lock. What did you lock?

_____

_____

2. Write your demo spine: task, input, tool chain, output, and one line on a hand-off.

_____

_____

_____

3. Name your two checks: one performance check and one governance/risk check.

Performance check	Governance / risk check

4. Prepare for a hard question: if asked how you know it is not misleading anyone, which check do you point to?

_____

_____

5. My reflection: I can show my system is trustworthy because... , low-risk and governance-aligned because... , and under a hard question I can point to...

_____

_____

Week 10 - Core - Worksheet | Showcase, Assessment & Reflection - Showcase & Reflection

Name: _____ **Class:** _____ **Date:** _____

Your brief: present and justify your solution, govern it on a six-part canvas, and set a Term-2 goal.

1. Justify two choices. For each, give the choice, its strand, the reason, and the trade-off.

Choice	Strand	Reason	Trade-off

2. Complete your governance canvas (fill each of the six parts).

Purpose & benefit	Risks & affected groups	Mitigations & owners
Data & privacy	Human oversight	Monitoring & review

3. Decision brief: do you recommend deploy, scale, or pause - and why?

_____

_____

4. Four-strand task: write one short, justified item from each strand.

_____

_____

_____

5. My reflection: the strongest choice I can justify, its strand and its trade-off, is... My Term-2 goal is...

_____

_____

Week 10 - Lab - Worksheet | Showcase, Assessment & Reflection - Peer Review & Aim Higher

Name: _____ **Class:** _____ **Date:** _____

Your brief: review others by strand, act on the best feedback, and set a prioritised Term-2 goal.

1. Review three solutions. For each, write one evidenced strength and one actionable improvement, tied to a strand.

Solution	Strand	Strength (evidence)	Improvement (actionable)

2. Make the highest-value fix from your own feedback. What did you change?

_____

_____

3. What did that fix ripple into - for better or worse - elsewhere in your system?

_____

_____

4. Write one prioritised Term-2 goal, the strand it targets, and a first step you can start on.

_____

_____

5. My reflection: the one Term-2 goal I am setting, the strand it targets, and my first step, is...

_____

_____